

KHA'SHUL, THE SYNTAX OF DEFINITION

A constructed language designed as the game's puzzle system: 10 words, 5 rules, one rule taught per puzzle.

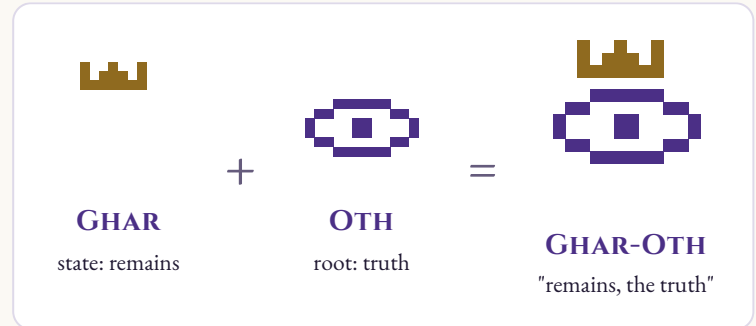
FROM THE MANUSCRIPT

"The Tribunal Gods speak in the Unification Tongue, a language of soft vowels and diplomatic nuance. But the Ancients spoke in **Kha'shul**. It was not a language for conversation. It was a syntax of definition. In Kha'shul, you did not ask a stone to move; you **redefined its coordinate**."

The book gives one word, **GHAR-OTH**, "the Truth that Remains", and three facts: it deals in absolutes ("it has no word for *friend*"), it sounds "sharp, guttural", and it is dangerous for mortal throats. Everything else here is designed from those.

STRUCTURE: STATE FIRST

A sentence is **STATE-ROOT**: a *state* (rises, holds, is still) then a *root* (stone, mist, fire). Saying it makes the state true of the thing. Order carries no extra meaning: there is no tense, mood or question, and any other order is not a sentence.



DESIGN GOALS

- **Reading is gameplay.** The player learns real grammar, not passwords.
- **Small and exact.** 10 words; one legal word order; no exceptions.
- **Alien but learnable.** The definition comes *before* the thing it defines.
- **Failure is content.** Wrong but grammatical sentences do something harmless and get a reaction.
- **Designer-owned data.** Words live in `Lexicon.txt`, puzzles in `Definitions.txt`.

GLYPHS: SHAPE AND STROKE

Every **root** is a **shape** and every **state** is a **stroke** added to it, so a spoken sentence is written as one glyph: the root's shape carrying the state's stroke (a compound's second root stands beside it). The game draws each glyph from a 16 x 16 pattern (`Glyphs.cs`); the images in this document are those same patterns.

SOUND

Closed syllables and hard consonants, nothing soft: *Ghar, Kel, Tor, Mur, Rhen; Oth, Duun, Vash, Iss, Sacl*. The voice-over pins each word's pronunciation.



Lyssia's notebook at the fallen bridge: every copied word as a card; the sentence is written as one glyph (left) and spoken on Enter.

THE LEXICON

Five states, five roots. States are drawn in gold, roots in violet, as in Lyssia's notebook.

GLYPH	WORD	KIND	MEANING	GLYPH ID	LEARNED IN
	GHAR	STATE	remains, endures, is unveiled	crown	The Golden Drift (area 2)
	KEL	STATE	rises, lifts	rising	The Golden Drift (area 2)
	TOR	STATE	holds, is solid	bar	The Throat (area 3)
	MUR	STATE	is still, is silent	underline	The Throat (area 3)
	RHEN	STATE	turns, orbits, returns	orbit	The Sanctuary (area 5)
	OTH	ROOT	truth, what is revealed	eye	The Golden Drift (area 2)
	DUUN	ROOT	stone, the heavy	block	The Golden Drift (area 2)
	VASH	ROOT	mist, breath	spiral	The Throat (area 3)
	ISS	ROOT	fire, heat	flame	The Throat (area 3)
	SAEL	ROOT	the self, the speaker	circle	The Throat (area 3)

Why these ten. Each puzzle needs one new idea and at most two new words. The states pair up as opposites the player can reason with (*rises* / *is still*, *holds* / *turns*), and the words are the things the chapter is made of: stone, mist, fire, truth, and the self.

Where words come from. Words are copied from carvings and from the Black Mirror's glyph, in the order the puzzles need them. The notebook only ever shows words Lyssia has copied, so the choice space grows one step at a time: 4 words at the bridge, 8 at the mist, 9 at the Mirror, 10 in the Sanctuary.

GRAMMAR: FIVE RULES

Taught in a teacher's order: introduce, practise, combine. Each rule is carved at the puzzle that needs it.

RULE	STATEMENT		EXAMPLE	IN PLAY
1. Declare	A sentence is STATE-ROOT. It makes the state true.		KEL-DUUN <i>rises, the stone</i>	The fallen bridge stands up.
2. No negation	There is no "not". To stop something, declare its opposite.		MUR-ISS <i>is still, the fire</i>	You can't tell fire to stop burning; you tell it what it is instead.
3. Compounds	Two roots joined are one new thing, which takes the state.	 	TOR-VASH-DUUN <i>holds, the mist-stone</i>	Tor-Vash (mist that holds) is too weak: 'like wet paper'.
4. Self-definition	Sael puts the speaker into the sentence. Some doors ask what is.		GHAR-OTH <i>remains, the truth</i>	Ghar-Sael (I remain) is refused: nobody survives the fire.
5. Coordinates	A state applied to the self redefines where things are around the speaker.		RHEN-SAEL <i>turns, to the self</i>	Each ring of books swings its gap round to Lyssia.

NOT SENTENCES

The parser (`KhashulParser`, pure C#, tested) rejects anything that isn't a sentence, and Lyssia explains why in one line. After the Touch, her explanations turn into Conceptual Bleed metaphors (right).

MISTAKE	EXAMPLE	LYSSIA	AFTER THE TOUCH
Root before state	DUUN-KEL	Backwards. In Kha'shul the definition comes first.	<i>The thing before its law? The stone laughs at me.</i>
Only roots	DUUN	Only names. Nothing is being defined.	<i>Nouns, drifting. Nothing holds them.</i>
Two states	KEL-MUR-DUUN	Two definitions for one thing. Kha'shul allows one.	<i>A thing wearing two truths wears neither.</i>
Only a state	KEL	A definition of... nothing. It needs something to hold.	<i>A law with no body. It falls through the floor.</i>
Three roots	KEL-DUUN-VASH-ISS	Too many things at once. Two can be joined, never three.	<i>Three things braided become a knot, not a word.</i>



Rule 1: Kel-Duun, and the bridge stands up.



Rule 2: the fire can only be declared still.

EVERY PUZZLE, EVERY ANSWER

From `Definitions.txt`. Wrong but grammatical sentences are designed, not errors.

THE FALLEN BRIDGE, THE JUNGLE 1. DECLARE

	SENTENCE	WHAT HAPPENS	REACTION
SOLVES	KEL-DUUN	the bridge stands up	RYSTAK ...The stone just stood up.
EFFECT	MUR-DUUN	the stone's hum stops	RYSTAK Great. Now it's quiet and broken.
EFFECT	KEL-ISS	nothing visible	CAELUS I asked for less fire, scholar. My face paint is melting.

THE FIRE, THE THROAT 2. NO NEGATION

	SENTENCE	WHAT HAPPENS	REACTION
SOLVES	MUR-ISS	the lava goes still	RYSTAK ...It went quiet. It's still hot. But it's quiet.
EFFECT	KEL-ISS	nothing visible	CAELUS I asked for less fire, scholar. My face paint is melting.
EFFECT	MUR-DUUN	the stone's hum stops	RYSTAK Great. Now it's quiet and broken.

THE MIST GAP, THE THROAT 3. COMPOUNDS

	SENTENCE	WHAT HAPPENS	REACTION
SOLVES	TOR-VASH-DUUN	the mist becomes a walkable floor	VESPER Mist-stone. Is that... a thing?
EFFECT	TOR-VASH	nothing visible	RYSTAK It's holding. Like wet paper.
EFFECT	KEL-VASH	nothing visible	VESPER Now it's in my face. Helpful.

EVERY PUZZLE, EVERY ANSWER (CONTINUED)

From `Definitions.txt`. Wrong but grammatical sentences are designed, not errors.

THE BLACK MIRROR 4. SELF-DEFINITION

	SENTENCE	WHAT HAPPENS	REACTION
SOLVES	GHAR-OTH	the glyph turns white; the mirror liquefies	LYSSIA Ghar-Oth. The truth that remains.
EFFECT	GHAR-SAEL	the glyph flares red	LYSSIA Ghar-Sael. I remain.
EFFECT	GHAR-DUUN	the glyph flares red	RYSTAK The stone remains. See? Steel, basically. I was right.
EFFECT	GHAR-ISS	the glyph flares red	VESPER "The fire remains"? That's not an answer. That's a threat.
OTHER	any other sentence	nothing	LYSSIA It isn't listening. It asked what survives. Something that remains.

THE ORBITS OF BOOKS, THE SANCTUARY 5. COORDINATES

	SENTENCE	WHAT HAPPENS	REACTION
ANSWER	RHEN-SAEL	the ring's gap swings round to Lyssia	LYSSIA Rhen-Sael. Turn, to the self.
EFFECT	RHEN-OTH	nothing visible	VESPER Now they're spinning faster and there are pages everywhere. Is that the truth? It looks like a mess.
EFFECT	RHEN-DUUN	nothing visible	CAELUS Only the crystal tablets twitched. The rest ignored you. Books have opinions.
OTHER	any other sentence	nothing	LYSSIA They turn around something. Not around that.

THE SENTINELS, THE ESCAPE REVIEW UNDER PRESSURE

	SENTENCE	WHAT HAPPENS	REACTION
ANSWER	MUR-DUUN	the nearest Sentinel freezes	a crew shout during the escape
ANSWER	TOR-DUUN	the nearest Sentinel freezes	a crew shout during the escape
ANSWER	KEL-DUUN	the nearest Sentinel is flung back	a crew shout during the escape
OTHER	any other sentence	nothing	LYSSIA It doesn't care.

The orbits and the Sentinels have no one-time solution: the ring can be turned again from anywhere, and the escape asks for the same sentences under time pressure (the notebook slows time to 15% while it is open).

DESIGN NOTES

FAILURE AS CONTENT

Every definition point lists the harmless sentences a curious player is likely to try, each with its own reaction. The table is small (two or three per point), but it covers the *near misses*: the right state on the wrong root (**MUR-DUUN** at the bridge silences the stone instead of lifting it), and the half-answer (**TOR-VASH** at the mist holds "like wet paper"). Near misses teach the rule better than a hint.

SENTENCES THAT WORK ANYWHERE

SENTENCE	WHAT HAPPENS	REACTION
TOR-SAEL	Lyssia is rooted to the spot	VESPER Stop defining yourself as furniture, scholar.

UNGRAMMATICAL ATTEMPTS

A grammar error never costs anything. Lyssia names the rule that was broken, which turns the notebook into a safe place to experiment.



The Black Mirror: its glyph shifts constantly until Ghar-Oth turns it white.

WHY NO "NOT"

"Kha'shul deals in absolutes." A language of definitions can't express absence, so Rule 2 turns a limitation into the puzzle: the fire can't be told to stop, only declared *still*. It also makes the chapter's theme playable: things are what they are defined to be.

THE MIRROR ASKS, THE CAMP ANSWERS

The Black Mirror's riddle, "What survives the fire?", is foreshadowed the night before: each companion answers it without being asked. At the Mirror their wrong answers call back what they said, and the player composes **GHAR-OTH** from two words learned at the Mouth an hour earlier. The last puzzle is solved with the first words the player ever copied.

CONCEPTUAL BLEED

After the Touch, the language starts to speak through Lyssia: 17 of her lines have a metaphor variant (**plain || bleed** in the script), and in the escape her Expression lengthens every stall.

BUILT TO GROW

Adding a word is one line in **Lexicon.txt** and a 16 x 16 pattern; adding a puzzle is a block in **Definitions.txt** and a dialogue node. Tests check every puzzle names a solution or an effect and that every reaction exists.